

ADS inputs as part of the display. Precision approach radars are used by military controllers, to assist the pilot in final phases of landing in places where instrument landing system and other sophisticated airborne equipment are unavailable to assist the pilots in marginal or near-zero visibility conditions. A radar archive system (RAS) is used to keep an electronic record of all the radar information for a few weeks, which can be useful for search and rescue. It is also useful for technicians maintaining the radar systems.

Space-based ATC

NASA's new space-based, higher-frequency, higher-bandwidth radio system can track aircrafts in real time across the globe by sending large quantities of data back and forth pretty fast. The communication system can be reprogrammed from a distance to ensure that flights are quickly located in case of

mishaps. The radios are used to create a space-based global air traffic control system.

Under this project, a constellation of 66 satellites equipped with the new radios will be launched into the orbit. The radios are programmed to receive signals from ADS-B airplane transceivers, which will automatically send out a flight's number, location, heading and other details. This will make it possible to keep track of all the aircrafts in the world within seconds. With real-time global tracking, planes will be able to fly with less space between them and therefore take more direct routes.

Coming up: airport cities

In a few years from now, biometric face recognition software, machine-readable passports and high-speed laser molecular scanners will enable fliers to clear security and customs checking at the airport within sec-

onds. Once inside the aircraft, they can look forward to being treated with massage seats where they are served various drinks and vitamins and provided a perfumed environment of their choice. Sound show-ers will help them relax and get a perfect night's sleep. The entire body of the plane will be made up of panoramic windows that can be made fully transparent at a click, thus providing travellers with astounding 360-degree views of the world around them.

Already we have some airports that are like a mini city, thus doing away with the need to step out into the city if you have a brief layover. Passenger processing technologies—such as those already in the marketplace—hold great promise to leverage greater system integration, mobile technology and smart devices for greater efficiency, lower operating costs and a transformative passenger experience. **EFY**



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